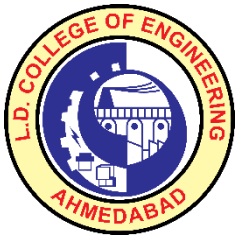

AN INTERNSHIP REPORT

ON

DATA ANALYTICS AND MACHINE LEARNING / DATA ANALYTICS USING POWER BI TOOL / MACHINE LEARNING BASED

MASTERS DECISION SUPPORT SYSTEM

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SEM 07

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WEEK 1/ Day 01	02 JULY 2025

Objective:

1. Types of Data Sources:
2. Dictionary Data Structure
3. Fetching data from live API , ISRO API , CRYPTO API

1. Types of Data Sources:

S.No	Data Source	Recommendation	Brief Explanation
1.	**Variable Data**	Not Recommended	Data hardcoded as variables in code (e.g., x = [1,2,3]). Not scalable or reusable for real projects.
2.	**Text Data (.txt)**	Not Recommended	Unstructured and lacks tabular format. Hard to analyze or visualize directly. Used only for simple data or logs.
3.	**CSV / Excel Data**	Recommended	Structured, easy to read, widely supported in Python (pandas). Ideal for tabular data (e.g., sales, customer records).
4.	**Folder Data**	Recommended	Useful for image/audio datasets where each file represents a sample (e.g., folders of cat/dog images for classification).
5.	**LiveData / API**	Recommended Most	Real-time, dynamic data pulled from web APIs (e.g., stock prices, weather). Great for real-world projects and automation.

2. Dictionary Data Structure:

A **dictionary** in Python is an **unordered collection of key-value pairs**, used to store data values like a map.

Key Features:

- Keys are **unique**
- To access value Key is required
- Defined using **curly braces {}**

```
In [1]: mydata = { "Ahmedabad": [{"date": "28 June 2025","cases": 15},
                                 {"date": "29 June 2025","cases": 45},
                                 {"date": "30 June 2025","cases": 35}
                                ],
                  "Surat": [300,350,1],
                 "Rajkot": 400
                 }
        # print all main keys
        print(mydata.keys())

        dict_keys(['Ahmedabad', 'Surat', 'Rajkot'])

In [2]: # print count of main keys

        print(len(mydata.keys()))

        3

In [3]: # print 400
        print(mydata["Rajkot"])

        400
```

3. Fetching Data from Live APIs:

API (Application Programming Interface) allows two applications to communicate.

In data science/ML, **APIs are used to fetch real-time data** from the internet, such as weather, financial, or scientific data.

How It Works:

1. You send a **request** (usually HTTP) to the API endpoint (URL).
2. The API sends back a **response**, usually in **JSON format**.
3. You parse this response in Python using libraries like requests.

API / JSON : LIVE / SERVER DATA <======> OFFLINE DATA : DICTIONARY

ISRO : https://isro.vercel.app/api/spacecrafts

In [6]:
```
import requests

url = requests.get(" https://isro.vercel.app/api/spacecrafts")
mydata = url.json()

# print(mydata)
print(type(mydata))
```

```
<class 'dict'>
```

In [7]:
```
# print main keys
print("Main keys in this isro API: ", mydata.keys())

# print number of main keys
print("Number of main keys: ", len(mydata.keys()))
```

```
Main keys in this isro API:  dict_keys(['spacecrafts'])
Number of main keys:  1
```

In [8]:
```
# print Aryabhat

print(mydata["spacecrafts"][0]["name"])
```

```
Aryabhata
```

CRYPTO : https://api.coinlore.net/api/tickers/

In [9]:
```
url = requests.get("https://api.coinlore.net/api/tickers/")
mydata = url.json()

print(mydata.keys())
print(type(mydata))
```

```
dict_keys(['data', 'info'])
<class 'dict'>
```

In [10]:
```
# print Bitcoin real time price

print("Real time", mydata["data"][0]["name"],"Price:" ,mydata["data"][0]["price_usd"])
```

```
Real time Bitcoin Price: 117543.77
```

WEEK 1/ Day 02	03 JULY 2024

Objective:

1. Printing data from API using loop
2. Search Operation on API
3. Pandas Data Frame
4. API data to CSV

1. Printing Data from API Using Loop

- When fetching data from an API, the response is often in **JSON format**, which contains **lists or dictionaries**. We use loops to iterate through this structured data and print specific values.

In [1]:
```
#api operations
import requests

url = requests.get("https://api.coinlore.net/api/tickers/")
mydata = url.json()

# print(mydata)
print(type(mydata))

# how many main keys are there ?
print(len(mydata.keys()))

#print all main keys
print(mydata.keys())

# allow user to insert key name, print found not found
userkey = input("Enter key name ")
for i in mydata:
    if userkey == i:
        print("key found")
        break
else:
    print("no such key in this api")

# how many crypto coins are there in this API ?
print("total number of coins: ", len(mydata["data"]))

# print name, sr no and price of all coins using for loop
for i in range(0,len(mydata["data"])):
    print( i+1,"Coin Name: ", mydata["data"][i]["nameid"],"Price: ",mydata["data"][i]["price_usd"])
```

2. Search Operation on API

- **Search operations** allow you to query specific data from an API using keywords, IDs, or parameters. Most APIs provide search endpoints where you can pass a search term to retrieve filtered results.

In [2]:
```
# Allow User to insert Coin Name, Print Real time price of that coin only(Search)

coinName = input("Enter coin name: ")
for i in range(len(mydata["data"])):
    if coinName == mydata["data"][i]["name"]:
        print("Current Price of ", coinName, " is ", mydata["data"][i]["price_usd"])
        break
else:
    print("coin not found")
```

```
Enter coin name: Bitcoin
Current Price of  Bitcoin  is  117932.06
```

3. Pandas DataFrame

- A **DataFrame** is the core data structure in the **pandas** library. It is a **2D labeled table** (like an Excel sheet) that holds **rows and columns**, making it ideal for data analysis and manipulation.

In [3]:
```
import pandas as pd

mydata = [15,25,35]
print(type(mydata))
pddata = pd.DataFrame(mydata)
print(pddata)
print(type(pddata))

newdata = (15,25,35)
print(type(newdata))
newpddata = pd.DataFrame(newdata)
print(newpddata)
print(type(newpddata))

impdata = {"Rohit": [75,12,124], "Virat": [87,45,92]}
print(type(impdata))
impdata = pd.DataFrame(impdata)
print(impdata)
print(type(impdata))

impdata.to_csv("testing.csv")
```

4. Saving API Data to CSV – In Brief

- After fetching data from an **API** (usually in **JSON format**), we often convert it into a **CSV file** for analysis or record-keeping. This is done using the pandas library.

In [4]:
```
# API to CSV

url = requests.get("https://api.coinlore.net/api/tickers/")
mydata = url.json()
print(type(mydata))
coindata = mydata["data"]
#print(coindata)
print(type(coindata))

pdcoindata = pd.DataFrame(coindata)
print(pdcoindata)

pdcoindata.to_csv("Livecoinprice.csv")
```

WEEK 1/ Day 03	04 JULY 2024

Objective:

1. Operations on Pandas DataFrame
2. Reading Data from CSV file
3. Multiple CSV file handling (RESULT ANALYSIS)

1. Operations on Pandas DataFrame

- Pandas provides powerful tools to perform a variety of operations on DataFrames to **analyze**, **clean**, and **manipulate** data efficiently.

In [1]:
```
import pandas as pd
import numpy as np

# Creating a DataFrame with performance scores of Rohit and Virat
impdata ={"Rohit": [75,12,124], "Virat": [87,45,92]}
impdata = pd.DataFrame(impdata, index=[1,2,3]) #customized index
print("Player Scores DataFrame:")
print(impdata)
print()

# Creating a DataFrame with player roles and scores in a particular game
game = pd.DataFrame([["Rohit","Batsman",50],["Kohli","Batsman",75]],
                    columns=["Name", "Type","Scores"], index =[1,2])
#customized index starts with 1
print("Game Performance DataFrame:")
print(game)
print()

# Creating a NumPy array of salaries over years for two employees
empdata = np.array([[8000,10000,18000],
                    [15000,18000,25000]])

# Converting NumPy array into a DataFrame with proper column and row labels
pdempdata = pd.DataFrame(empdata,
                         columns=[2022,2023,2024],
                         index=["Ramesh","Mahesh"])
print("Employee Salary DataFrame (2022-2024):")
print(pdempdata)
print()

# Adding a new column for 2025 with a 30% increment over 2024 salary
# pdempdata[2025]= (pdempdata[2024]*0.3) + pdempdata[2024]
pdempdata[2025] = pdempdata[2024] * 1.30
print("Employee Salary DataFrame (including 2025 with 30% increment):")
print(pdempdata)
```

📌 **.loc[] vs .iloc[] in Pandas**

.loc[] – Label-based indexing

- Used to access rows and columns by label names.
- Includes both start and end labels when slicing.
- **Syntax: df.loc[row_label, column_label]**

.iloc[] – Position-based indexing

- Used to access rows and columns by integer positions (like Python lists).
- End index is exclusive when slicing.
- **Syntax: df.iloc[row_position, column_position]**

2. Reading Data from CSV File

4. **CSV (Comma Separated Values)** is a common file format for storing tabular data. In Python, the pandas library provides a simple way to read CSV files into a **DataFrame** for analysis.

```
In [2]: # Reading data from a CSV file named "Livecoinprice.csv" into a DataFrame
        filedata = pd.read_csv("Livecoinprice.csv")

        # Printing the row at index 2 from the DataFrame using .loc (label-based indexing)
        # Note: .loc[[2]] returns the row where the index label is 2 (not necessarily the third row unless index is default)
        print(filedata.loc[[2]])
```

```
In [3]: filedata = pd.read_csv("mydata.xlsx - Sheet1.csv")
        print(filedata)
        print(type(filedata))
        print()

        # Print Score of all matches of Kohli
        print(filedata["KOHLI"])
        print()

        # Print 37
        # Specifica: Varname[Column][Row]
        print(filedata["DHONI"][3])
        print()

        # Print Third Match Data
        print(filedata.loc[[2]])
        print(filedata.iloc[[2]])
        # As the default index starts with 0, 3rd Match is in index 2
        # here label name and position both are 2 so both will print the same result
```

3. Handling Multiple CSV Files (Result Analysis)

5. In many real-world scenarios, data is split across **multiple CSV files** (e.g., results by subject, class, or semester). Using Python and pandas, you can **read, combine, and analyze** these files efficiently.

```
In [4]: df1 = pd.read_csv("RESULT1 - Sheet1.csv")
        print(df1)

        df2 = pd.read_csv("RESULT2 - Sheet1.csv")
        print(df2)

        alldata = pd.concat([df1,df2])
        alldata
```

```
In [5]: alldata.head(3)
```

Out[5]:

	SRNO	BRANCH	NAME	TOTAL	PERCENTAGE	PASSFAIL
0	1	CE	RAMESH	210	90	1
1	2	CE	SURESH	150	80	1
2	3	IT	MAHESH	225	75	1

```
In [6]: #sort data based on Total
        print(alldata.sort_values(['TOTAL'], ascending = False).head(3))
```

```
   SRNO BRANCH    NAME  TOTAL  PERCENTAGE  PASSFAIL
2     3     IT  KATHAN    285          95         1
1     2     CE   JATAN    270          90         1
2     3     IT  MAHESH    225          75         1
```

```
In [7]: print(alldata.sort_values(['TOTAL'], ascending = False).tail(3))
```

```
   SRNO BRANCH    NAME  TOTAL  PERCENTAGE  PASSFAIL
1     2     CE  SURESH    150          80         1
0     1     EC   RATAN    150          50         1
4     5     CE  JAYESH     90          30         0
```

```
In [8]: df1.dtypes
```

```
Out[8]: SRNO           int64
        BRANCH        object
        NAME          object
        TOTAL          int64
        PERCENTAGE     int64
        PASSFAIL       int64
        dtype: object
```

WEEK 1/ Day 04	07 JULY 2024

Objective:

1. Covid Data Analytics
2. Matplotlib : Bar Graph, Pie Chart, Line Graph
3. API data to Graph

1. COVID Data Analytics

It involves analyzing pandemic-related data to uncover patterns, trends, and insights about the spread and impact of the virus.

- Daily confirmed, discharged, and death cases
- State-wise case distribution

In [1]:
```
import requests
import pandas as pd

url = requests.get("https://api.rootnet.in/covid19-in/stats/latest")
mydata = url.json()

#print(mydata)
# print all main keys
print(mydata.keys())

# print total number of main keys
print("total number of keys: ", len(mydata.keys()))

# print Andaman and Nicobar islands
print(mydata['data']['regional'][0]['loc'])

# print total number of states in this API
print(len(mydata['data']['regional']))

# allow user to insert state name, Print state found not found
userinput = input("Enter state name: ")
for i in range(0,len(mydata['data']['regional'])):
    if userinput == mydata['data']['regional'][i]['loc']:
        print("State found")
        print("Total Confirmed: ", mydata['data']['regional'][i]['totalConfirmed'])
        print("Total Discharged: ", mydata['data']['regional'][i]['discharged'])
        print("Total Deaths: ", mydata['data']['regional'][i]['deaths'])
        break
else:
    print("State Not found")
```

```
dict_keys(['success', 'data', 'lastRefreshed', 'lastOriginUpdate'])
total number of keys:  4
Andaman and Nicobar Islands
36
Enter state name: Gujarat
State found
Total Confirmed:  1224657
Total Discharged:  1213502
Total Deaths:  10944
```

In [2]:
```
coviddata = pd.DataFrame(mydata['data']['regional'])
coviddata.head()
```

In [3]:
```
coviddata.to_csv("covid.csv")
```

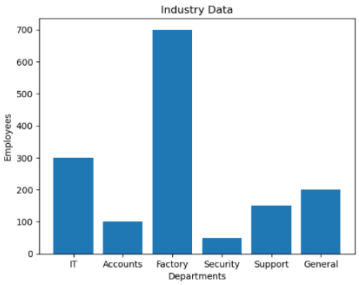
In [4]:
```
# Conditional data fetching:
covidpd = pd.read_csv("covid.csv")

# States having cases between 100000 to 150000
covidpd[(covidpd["totalConfirmed"]>=100000)& (covidpd["totalConfirmed"]<= 150000)]
```

2. Matplotlib – Bar Graph, Pie Chart, Line Graph (In Brief)

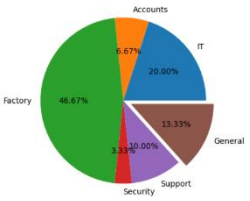
It is a Python library used for creating static, animated, and interactive plots. It is widely used for data visualization in data science and ML projects. Python code and it's respective Outputs:

```
In [6]: import matplotlib.pyplot as plt
```

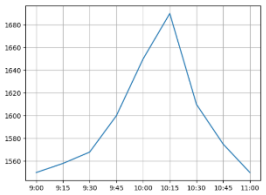
```
In [7]: # Bar Graph
departments = ["IT","Accounts","Factory", "Security", "Support", "General"]
employees = [300,100,700,50, 150,200]

plt.bar(departments,employees)
plt.title("Industry Data")
plt.xlabel("Departments")
plt.ylabel('Employees')
plt.show()
```




```
In [8]: # pie chart

plt.pie(employees, labels =departments, autopct="%.2f%%", explode=[0,0,0,0,0,0.1])
plt.show()
```

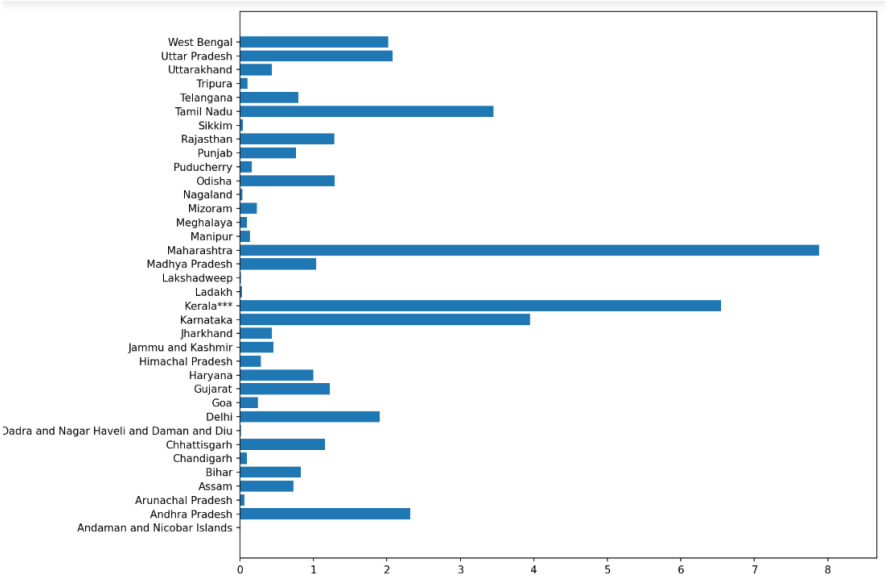



```
In [9]: # Line graph

time = ['9:00','9:15', '9:30', '9:45','10:00','10:15','10:30','10:45','11:00']
reliance = [1550,1558,1568,1600,1650,1690,1610,1575,1550]

plt.plot(time, reliance)
plt.grid(True)
plt.show()
```



3. API Data to Graph: [CovidDataset]

```
In [10]: # horizontal Bar Chart
covidpd = pd.read_csv("covid.csv")
plt.barh(covidpd['loc'],covidpd['totalConfirmed'])
plt.title("Covid Cases Analysis")
# plt.xlabel("Covid Cases")
# plt.ylabel("States")
plt.show()
```



WEEK 1/ Day 05	08 JULY 2024

Objective:

1. Machine Learning : Linear Regression
   (Mathematics + Coding)

What is Linear Regression?

Linear Regression is the **simplest form** of supervised learning used for predicting continuous numerical values.

Types of Linear Regression

- **Simple Linear Regression**: 1 independent variable
- **Multiple Linear Regression**: 2 or more independent variables

We aim to find the **best-fit straight line** through the data points.

Mathematical Model

For **Simple Linear Regression**, the model predicts output $\hat{y}$ using:

$$[\ \hat{y} = \beta_0 + \beta_1 x\ ]$$

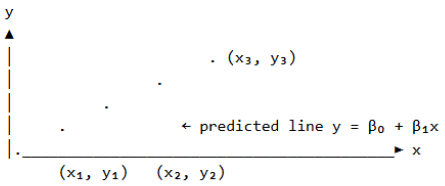
Where:

- $\hat{y}$: Predicted value (output)
- x: Input feature
- β_0: Intercept (bias)
- β_1: Slope (weight)

Graphical Interpretation

- The X-axis represents the input feature x
- The Y-axis represents the predicted output $\hat{y}$
- The best-fit line: $y = \beta_0 + \beta_1 x$
- Each point contributes a vertical error (called **residual**) to the line

Example visualization:



Data -> Visualization Data Pattern/ Data Nature Helps ML Algorithm

Data -> Model -> Train -> Predict

EQUATION OF SLOPE $\hat{\beta}_1 = \frac{\sum(x_i-\bar{x})(y_i-\bar{y})}{\sum(x_i-\bar{x})^2}$

	x	y	$(x-\bar{x})$	$(y-\bar{y})$	$(x-\bar{x})^2$
	1	2			
	2	4			
	3	5			
	4	4			
	5	5			
Σ					
	$\bar{x}$ =	$\bar{y}$ =			

	x	y	$(x-\bar{x})$	$(y-\bar{y})$	$(x-\bar{x})^2$	$(x-\bar{x})(y-\bar{y})$
	1	2	-2	-2	4	4
	2	4	-1	0	1	0
	3	5	0	1	0	0
	4	4	1	0	1	0
	5	5	2	1	4	2
Σ	15	20			10	6
	$\bar{x} = 3$	$\bar{y} = 4$				

$$\hat{\beta}_1 = \frac{\sum(x_i-\bar{x})(y_i-\bar{y})}{\sum(x_i-\bar{x})^2}$$

$= 6 / 10$

$B_1 = 0.6$

Put all values in slope equation ($\bar{y}$, B_1, X mean)

$\bar{y} = B_0 + B_1 X$

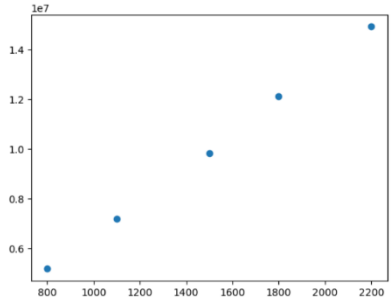
In [1]:
```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("areaPriceDay5.csv")
df
```

Out[1]:

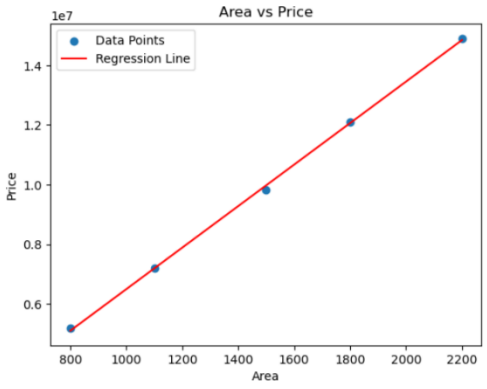
	area	price
0	800	5180000
1	1100	7190000
2	1500	9820000
3	1800	12100000
4	2200	14900000

In [2]:
```
plt.scatter(df['area'],df['price'])
```

Out[2]: <matplotlib.collections.PathCollection at 0x26c5cdddcf0>



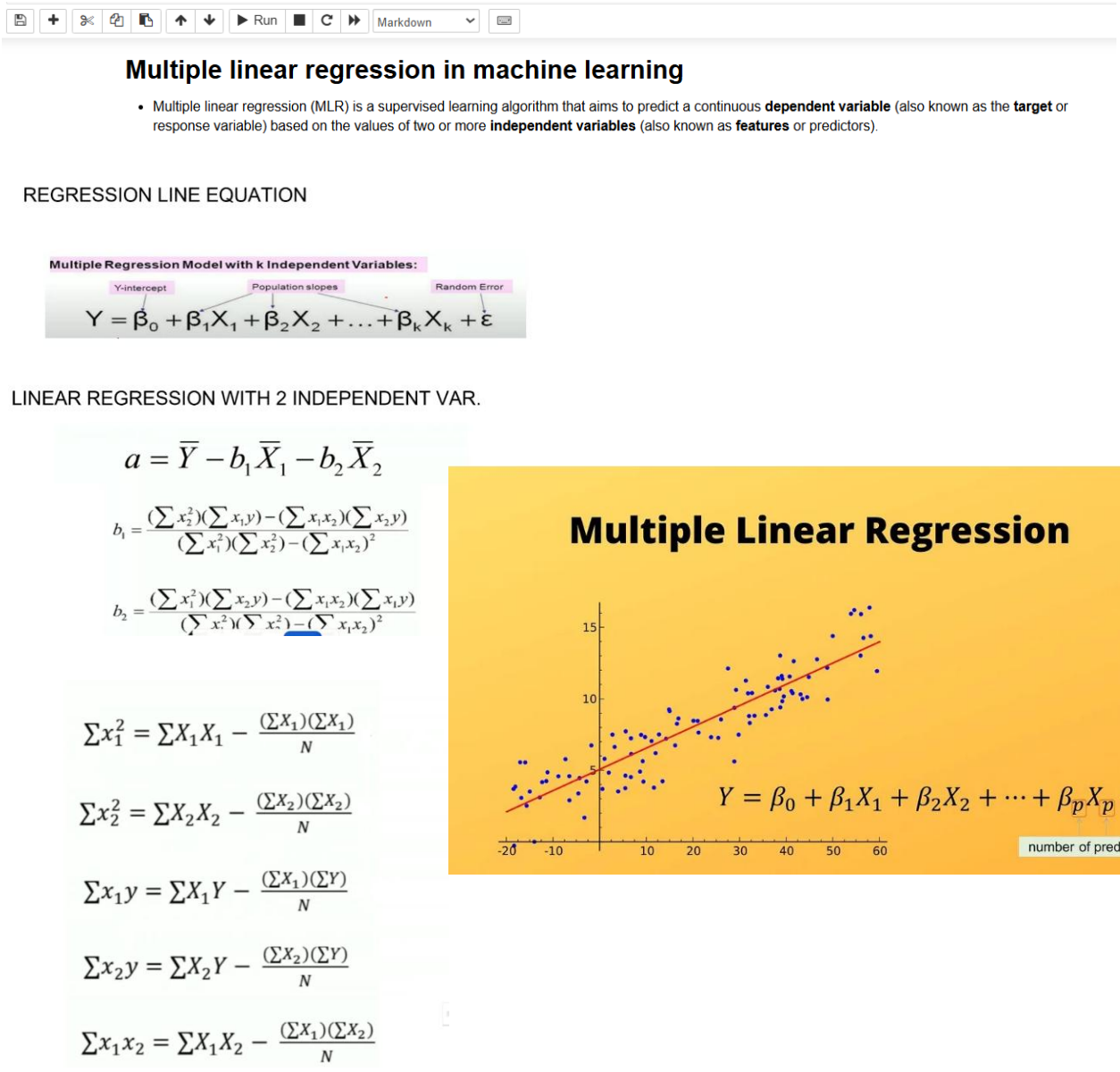
In [3]:
```
import matplotlib.pyplot as plt
import numpy as np

# Scatter plot
plt.scatter(df['area'], df['price'], label='Data Points')

# Fit a straight line (linear regression)
m, b = np.polyfit(df['area'], df['price'], 1)  # y = mx + b

# Plot the regression line
plt.plot(df['area'], m * df['area'] + b, color='red', label='Regression Line')

# Labels and legend
plt.xlabel('Area')
plt.ylabel('Price')
plt.title('Area vs Price')
plt.legend()
plt.show()
```



WEEK 1/ Day 06	09 JULY 2024

Objective:

1. Machine Learning : Multiple Linear Regression
   (Mathematics + coding)



1. The general form of the multiple linear regression model
2. Manual formula derivation and calculation steps
3. Step-by-step summation and coefficient calculation
4. Final equation interpretation and graph visualization

	INDEPENDENT		DEPENDENT					
	X_1	X_2	Y	X_1X_1	X_2X_2	X_1X_2	X_1Y	X_2Y
	3	8	-3.7	9	64	24	-11.1	-29.6
	4	5	3.5	16	25	20	14	17.5
	5	7	2.5	25	49	35	12.5	17.5
	6	3	11.5	36	9	18	69	34.5
	2	1	5.7	4	1	1	11.4	5.7
Σ	20	24	19.5	90	148	99	95.8	45.6

$$b_1 = \frac{(\sum x_2^2)(\sum x_1 y) - (\sum x_1 x_2)(\sum x_2 y)}{(\sum x_1^2)(\sum x_2^2) - (\sum x_1 x_2)^2} = \frac{32.8 * 17.8 - 3 * (-48)}{10 * 32.8 - 3 * 3} = 2.28$$

$$b_2 = \frac{(\sum x_1^2)(\sum x_2 y) - (\sum x_1 x_2)(\sum x_1 y)}{(\sum x_1^2)(\sum x_2^2) - (\sum x_1 x_2)^2} = \frac{10 * (-48) - 3 * 17.8}{10 * 32.8 - 3 * 3} = -1.67$$

$$a = \overline{Y} - b_1 \overline{X}_1 - b_2 \overline{X}_2 = \frac{19.5}{5} - \frac{2.28 * 20}{5} - \frac{-1.67 * 24}{5} = 2.79$$

Final Regression equation or Model is:

$$Y = 2.796 + 2.28\,x_1 - 1.67x_2$$

Multiple Linear Regression Jupyter Notebook Code:

Dataset: Boston Housing (predict median home value from multiple features)

```
In [1]: from sklearn.datasets import fetch_california_housing
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import root_mean_squared_error
import numpy as np
import pandas as pd

# Load data
california = fetch_california_housing()
X_full = pd.DataFrame(california.data, columns=california.feature_names)
y_full = pd.Series(california.target, name='MedHouseValue')

# Select 2 features for MLR
X = X_full[['MedInc', 'AveRooms']]
y = y_full
print(X)

# Split into train and test
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)

# Fit model
model = LinearRegression()
model.fit(X_train, y_train)

# Predict and evaluate
y_pred = model.predict(X_test)
rmse = root_mean_squared_error(y_test, y_pred)

# Results
print(f"Intercept: {model.intercept_:.4f}")
print("Coefficients:", dict(zip(X.columns, model.coef_)))
print(f"RMSE: {rmse:.4f}")
```

```
       MedInc  AveRooms
0      8.3252  6.984127
1      8.3014  6.238137
2      7.2574  8.288136
3      5.6431  5.817352
4      3.8462  6.281853
...       ...       ...
20635  1.5603  5.045455
20636  2.5568  6.114035
20637  1.7000  5.205543
20638  1.8672  5.329513
20639  2.3886  5.254717

[20640 rows x 2 columns]
Intercept: 0.5973
Coefficients: {'MedInc': 0.4362608864728618, 'AveRooms': -0.04017160872638347}
RMSE: 0.8379
Predicted MedHouseValue: 3.430439780418422
```

- Output

```
# Define the input data
input_data = [[7.2574, 8.288136]]

# Convert to DataFrame with the same column names as the training data
input_df = pd.DataFrame(input_data, columns=['MedInc', 'AveRooms'])

# Make the prediction
y_pred = model.predict(input_df)
print("Predicted MedHouseValue:", y_pred[0])
```

WEEK 2/ Day 07	10 JULY 2024

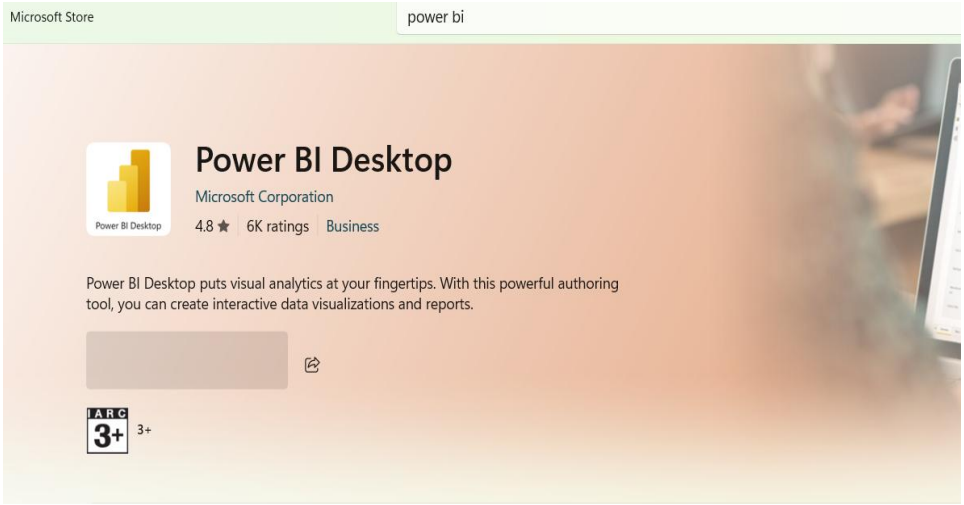
Objective:

1. PowerBI setup
2. Basics of Power BI

1. Power BI Setup:

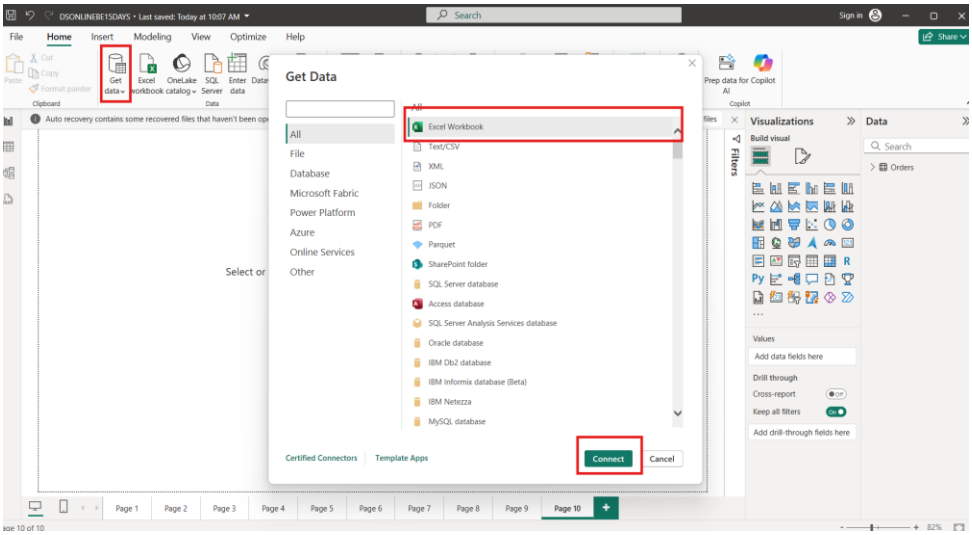
Power BI is a business analytics tool by Microsoft used to create interactive visualizations and dashboards.

Step 1 : Download Power BI Desktop from Microsoft Store.



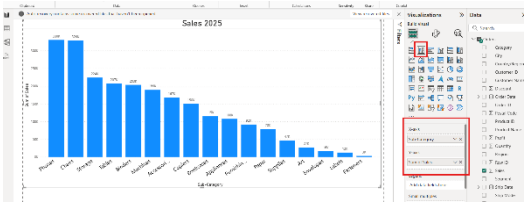
Step 2 : Load xlsx file in power bi.

Download Sample Superstore file: [Click here]

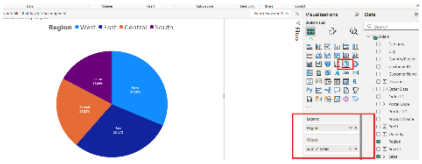


2. Basics of Power BI:

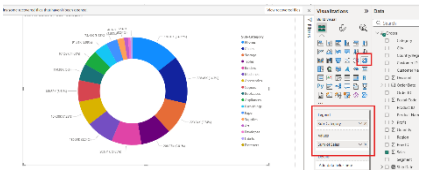
1. Select **Bar Chart** From Visuals
   - X Axis - Sub Category
   - Y Axis – Sales



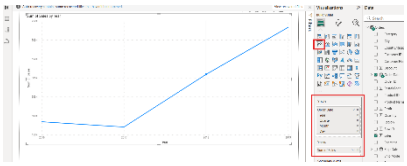
2. Select **Pie Chart** From Visuals
   - Legend : Region
   - Values: Sales



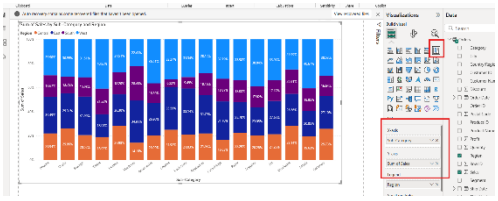
3. Select **Donut Chart** From Visuals
   - Legend : Sub Category
   - Values: Sales



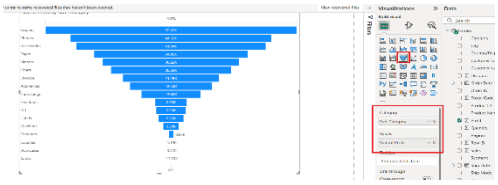
4. Select **Line Chart** From Visuals
   - X Axis - Order Date
   - Y Axis – Sales



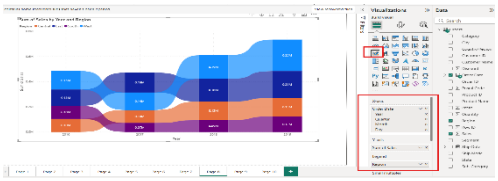
5. Select **Stack Bar Chart** From Visuals
   - X Axis - Sub Category
   - Y Axis - Sales
   - **Legend – Region**



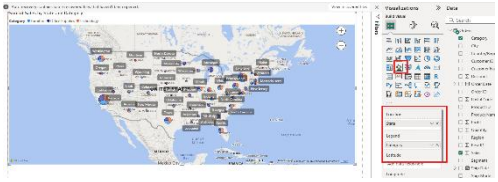
6. Select **Funnel Chart** From Visuals
   - Category : Sub-Category
   - Values: Profit



7. Select **Ribbon Chart** From Visuals
   - X Axis - Order Date
   - Y Axis – Sales



8. Select **Map Chart** From Visuals
   - Location: State
   - Legend – Category

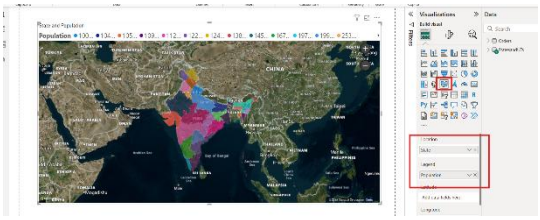


WEEK 2/ Day 08	11 JULY 2024

Objective:

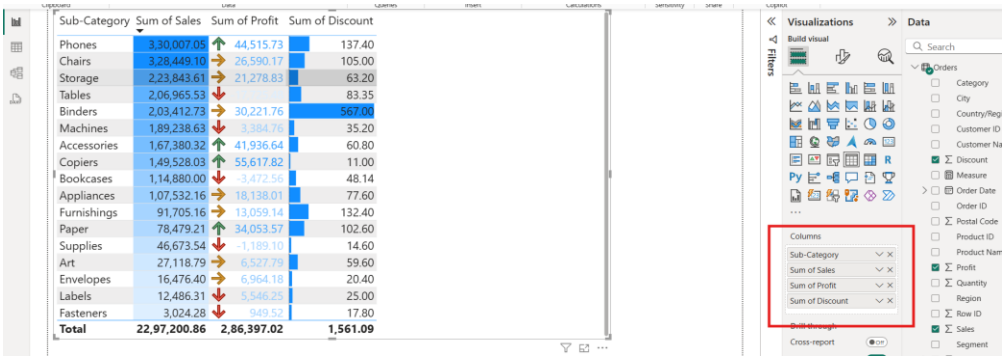
1. Graphs and Charts in PowerBI (Continued from yesterday)

Download India xlsx file: Click Here



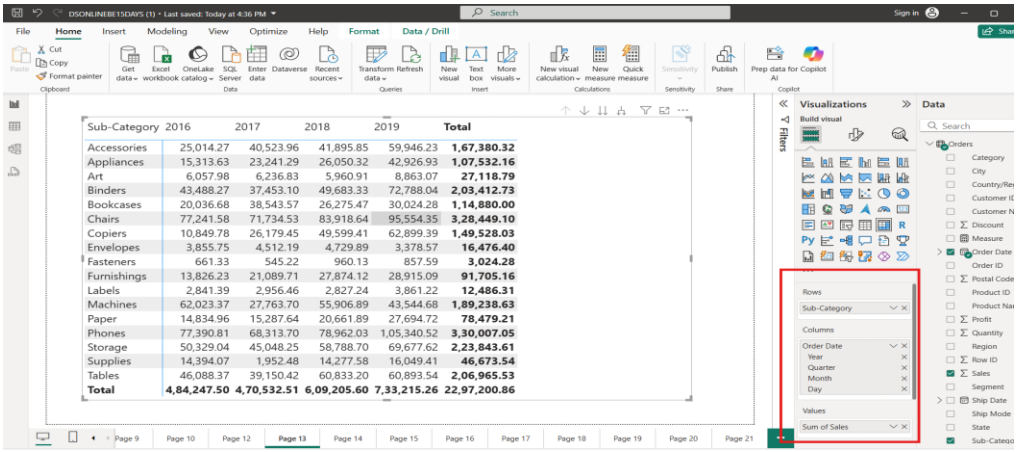
9. Select **Map Chart** From Visuals
   - Location: State
   - Legend – Population

10. Select **Table Chart** From Visuals
    - Columns: subcategory , sales , profit , discount



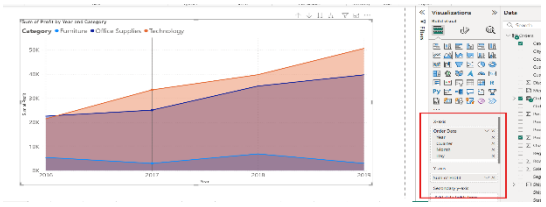
Sub-Category	Sum of Sales	Sum of Profit	Sum of Discount
Phones	3,30,007.05	44,515.73	137.40
Chairs	3,28,449.10	26,590.17	105.00
Storage	2,23,843.61	21,278.83	63.20
Tables	2,06,965.53		83.35
Binders	2,03,412.73	30,221.76	567.00
Machines	1,89,238.63	-3,384.76	35.20
Accessories	1,67,380.32	41,936.64	60.80
Copiers	1,49,528.03	55,617.82	11.00
Bookcases	1,14,880.00	-3,472.56	48.14
Appliances	1,07,532.16	18,138.01	77.60
Furnishings	91,705.16	13,059.14	132.40
Paper	78,479.21	34,053.57	102.60
Supplies	46,673.54	-1,189.10	14.60
Art	27,118.79	6,527.79	59.60
Envelopes	16,476.40	6,964.18	20.40
Labels	12,486.31	5,546.25	25.00
Fasteners	3,024.28	949.52	17.80
Total	**22,97,200.86**	**2,86,397.02**	**1,561.09**

11. Select **Matrix Chart** From Visuals
    - Rows : Sub Category
    - Columns : Order Date
    - Values: Sales

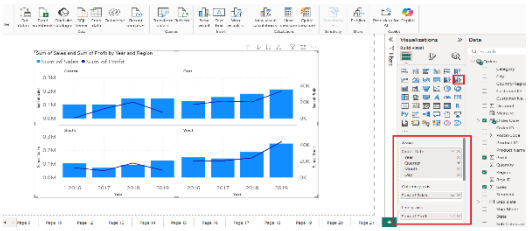


Sub-Category	2016	2017	2018	2019	Total
Accessories	25,014.27	40,523.96	41,895.85	59,946.23	**1,67,380.32**
Appliances	15,313.63	23,241.29	26,050.32	42,926.93	**1,07,532.16**
Art	6,057.98	6,236.83	5,960.91	8,863.07	**27,118.79**
Binders	43,488.27	37,453.10	49,683.33	72,788.04	**2,03,412.73**
Bookcases	20,036.68	38,543.57	26,275.47	30,024.28	**1,14,880.00**
Chairs	77,241.58	71,734.53	83,918.64	95,554.35	**3,28,449.10**
Copiers	10,849.78	26,179.45	49,599.41	62,899.39	**1,49,528.03**
Envelopes	3,855.75	4,512.19	4,729.89	3,378.57	**16,476.40**
Fasteners	661.33	545.22	960.13	857.59	**3,024.28**
Furnishings	13,826.23	21,089.71	27,874.12	28,915.09	**91,705.16**
Labels	2,841.39	2,956.46	2,827.24	3,861.22	**12,486.31**
Machines	62,023.37	27,763.70	55,906.89	43,544.68	**1,89,238.63**
Paper	14,834.96	15,287.64	20,661.89	27,694.72	**78,479.21**
Phones	77,390.81	68,313.70	78,962.03	1,05,340.52	**3,30,007.05**
Storage	50,329.04	45,048.25	58,788.70	69,677.62	**2,23,843.61**
Supplies	14,394.07	1,952.48	14,277.58	16,049.41	**46,673.54**
Tables	46,088.37	39,150.42	60,833.20	60,893.54	**2,06,965.53**
Total	**4,84,247.50**	**4,70,532.51**	**6,09,205.60**	**7,33,215.26**	**22,97,200.86**

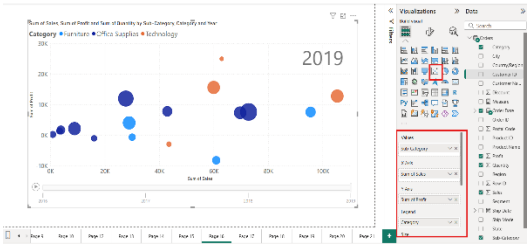
12. Select **Area Chart** From Visuals
    - X Axis : Order Date
    - Y Axis : Profit
    - Legend: Category



13. Select **Line and Column Chart**
    - X Axis: Order Date
    - Column Y Axis- Sales
    - Line Y Axis: Profit
    - Small Multiple: Region

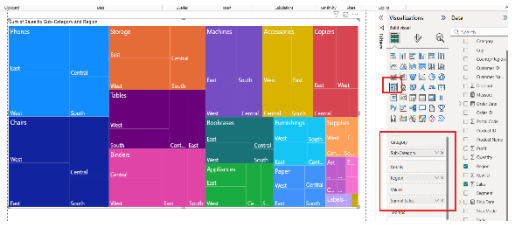


14. Select **Scatter Chart** From Visuals
    - Values: SubCategory
    - X Axis: Sales
    - Y Axis: Profit
    - Legend: Category
    - Size: Quantity
    - Play Axis: Order Date (Date Hierarchy)

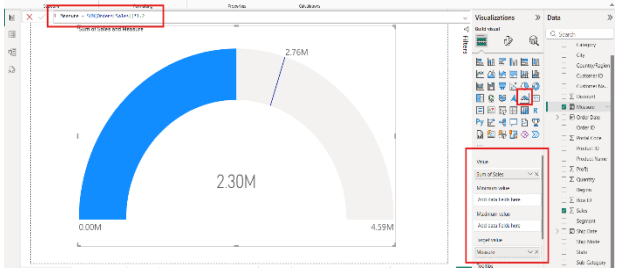


15. Select **WaterFall Chart** From Visuals
    - Category: Order Date
    - Breakdown: Region
    - Y Axis- Sales

16. Select **Tree Chart** From Visuals
    - Category: Subcategory
    - Value- Sales
    - Details: Region



17. Select Gauge Chart From Visuals
    - Value: Sales
    - Target Value: Measure = `SUM(Orders[Sales])*1.2`



WEEK 2/ Day 09	14 JULY 2024

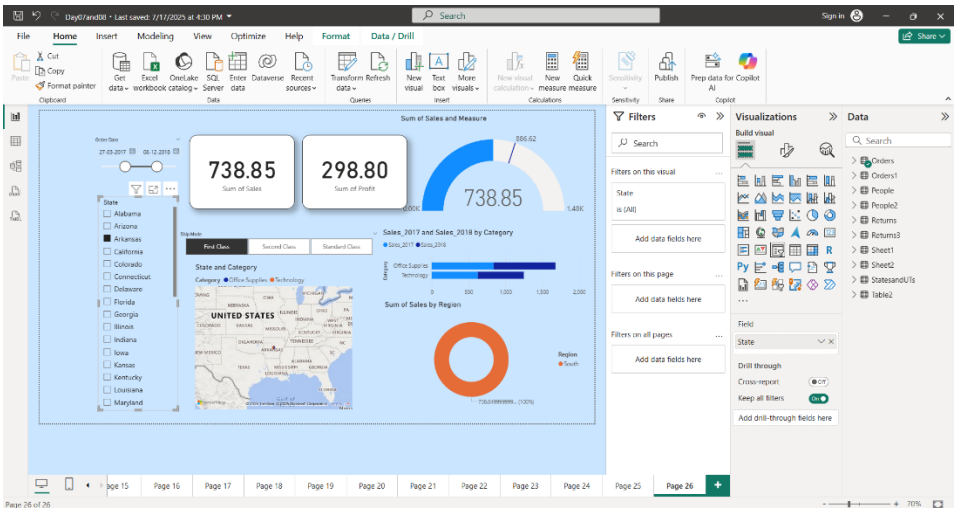
Objective:

1. Data Handling : Advance Power BI **(Project: Dashboard Design)**

1. Project: Dashboard Design

The main objective of creating this Power BI dashboard was to analyze sales and profit performance based on several business dimensions such as **time period, state-wise sales, shipping method, product categories, and regional distribution**.

- This report aims to provide meaningful insights to support business decisions.



2. Dashboard Design Highlights

Component	Purpose
Date Slicer	Filter data dynamically by order date
State Slicer	Focus on sales by individual states
Ship Mode Buttons	Segment sales/profits by delivery method
KPI Cards (Sales/Profit)	Quick summary of total sales and profit
Gauge Chart	Visualization of current sales against performance target
Bar Chart	Year-wise and category-wise comparison of sales
Map Visualization	Geographic distribution of sales in the United States
Donut Chart	Regional sales contribution in percentage

WEEK 2/ Day 10	15 JULY 2024

Objective:

1. Random Forest project : Master Degree **opt**/**not opt** Prediction

Masters Decision Support System:

Project Overview

This Streamlit-based project predicts whether a student should pursue a Master's degree based on their GATE score and current salary using a machine learning model.
It features a clean user interface with data visualization, prediction form, and dataset preview.

Features in the Application

- Home: Welcome screen and navigation
- About Model: Describes model type and purpose
- Dataset: Shows the contents of **student_career_data.csv**
- Prediction Model: Predicts if a user should pursue a Master's
- Graph: Visualizes salary, GATE score, and decision distribution

Machine Learning Model Details

- Model Type: Random Forest Classifier
- Features Used: GATE Score (Categorical), Salary (Numerical)
- Target Column: Should_Do_Masters
- Data Preprocessing: Label Encoding of categorical columns
- Train-Test Split: 80% training, 20% testing

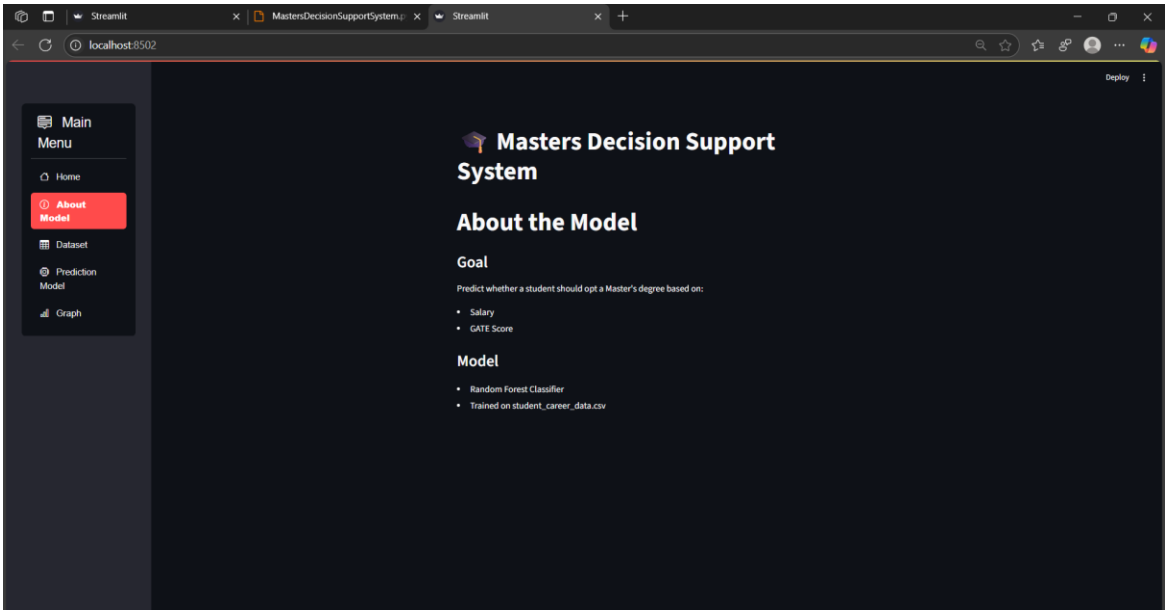
Code

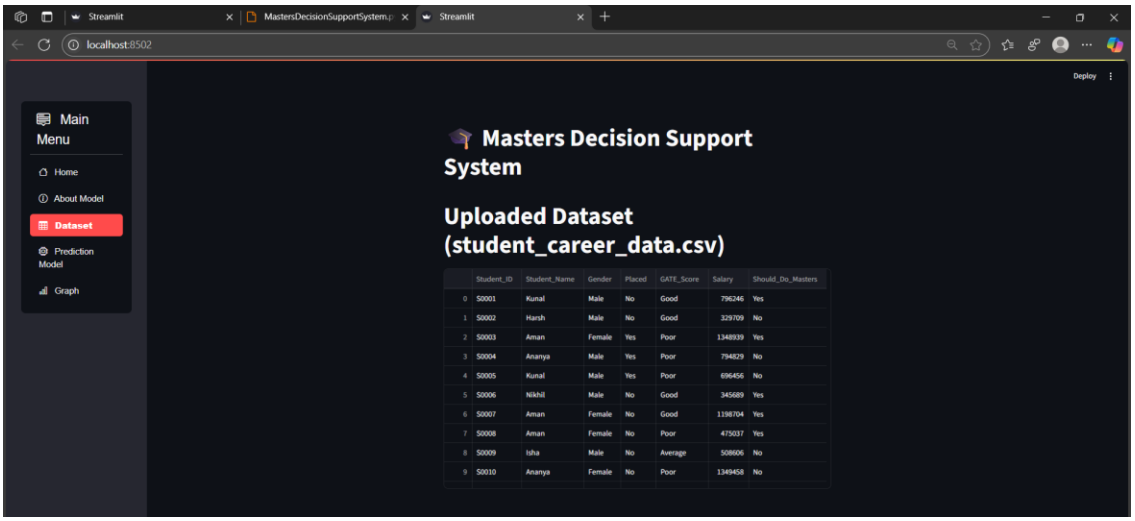
```python
import streamlit as st
from streamlit_option_menu import option_menu
from sklearn.ensemble import RandomForestClassifier
from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
import pandas as pd
import matplotlib.pyplot as plt

# Load and prepare data
model_df = pd.read_csv("student_career_data.csv")

le_gate = LabelEncoder()
model_df["GATE_Score"] = le_gate.fit_transform(model_df["GATE_Score"])

le_master = LabelEncoder()
model_df["Should_Do_Masters"] = le_master.fit_transform(model_df["Should_Do_Masters"])

if le_master.transform(['Yes'])[0] != 1:
    model_df["Should_Do_Masters"] = 1 - model_df["Should_Do_Masters"]
    le_master = LabelEncoder()
    le_master.fit(["No", "Yes"])

gate_label_map = dict(zip(le_gate.transform(le_gate.classes_), le_gate.classes_))
master_label_map = {0: "No", 1: "Yes"}
```

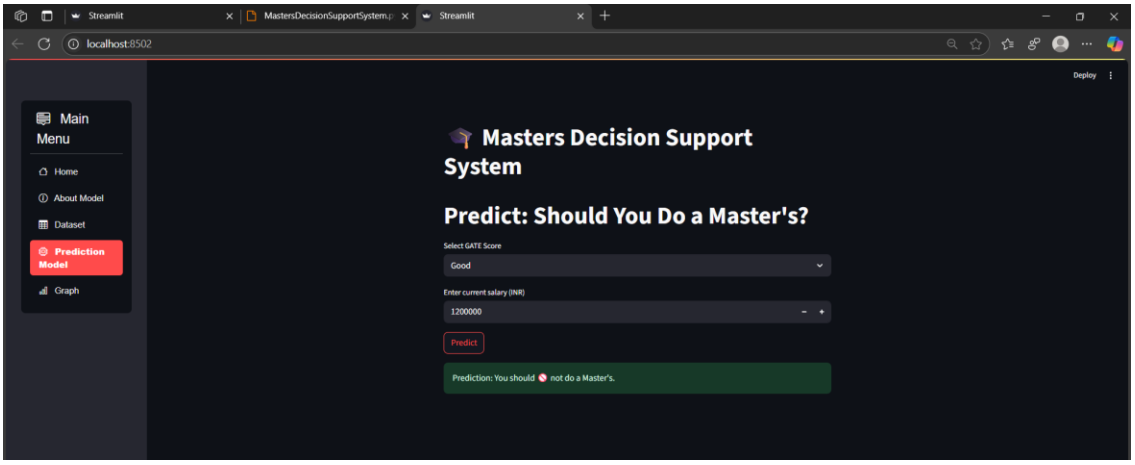
```python
X = model_df[["GATE_Score", "Salary"]]
y = model_df["Should_Do_Masters"]

x_train, x_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
model = RandomForestClassifier(n_estimators=500, random_state=42)
model.fit(x_train, y_train)

# UI
st.title("🎓 Masters Decision Support System")

with st.sidebar:
    selected = option_menu(
        menu_title="Main Menu",
        options=["Home", "About Model", "Dataset", "Prediction Model", "Graph"],
        icons=["house", "info-circle", "table", "cpu", "bar-chart"],
        orientation="vertical"
    )

# Pages
if selected == "Home":
    st.subheader("👋 Welcome!")
    st.write("View Main Menu from sidebar.")

elif selected == "About Model":
    st.title("📘 About the Model")
    st.markdown("""
    ### 🎯 Goal
    Predict whether a student should opt a Master's degree based on:
    - Salary
    - GATE Score

    ### 🧠 Model
    - Random Forest Classifier
    - Trained on student_career_data.csv


    """)

elif selected == "Dataset":
    st.title("📊 Uploaded Dataset (student_career_data.csv)")
    try:
        dataset_df = pd.read_csv("student_career_data.csv")
        st.dataframe(dataset_df)
    except FileNotFoundError:
        st.error("data.csv file not found. Please upload or check path.")
```

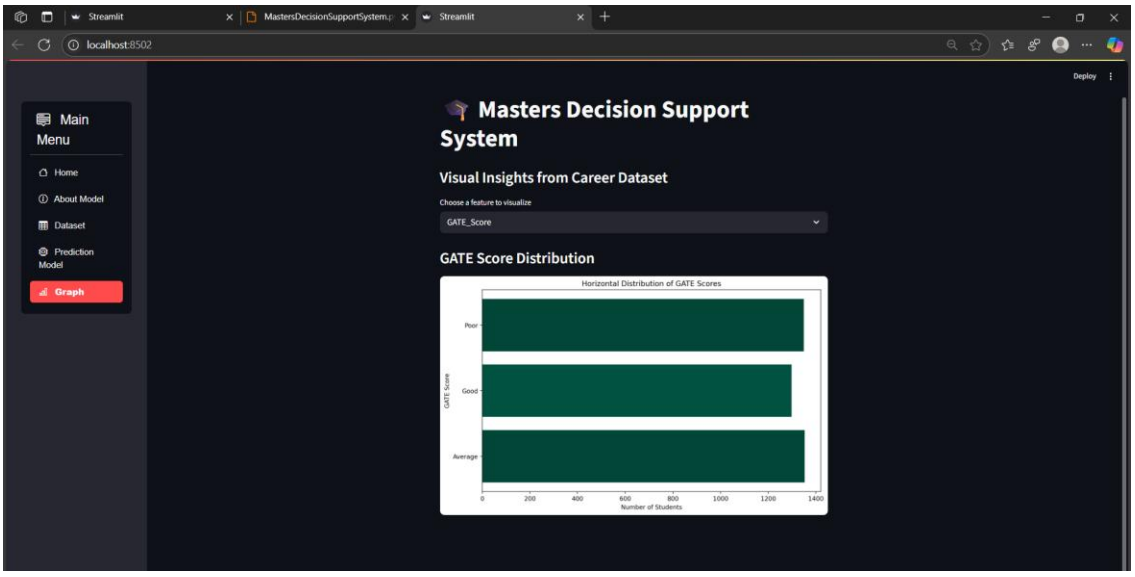
```python
elif selected == "Prediction Model":
    st.title("🔮 Predict: Should You Do a Master's?")
    gate_input = st.selectbox("Select GATE Score", le_gate.classes_)
    salary_input = st.number_input("Enter current salary (INR)", value=500000)

    if st.button("Predict"):
        gate_encoded = le_gate.transform([gate_input])[0]
        prediction = model.predict([[gate_encoded, salary_input]])
        result = le_master.inverse_transform(prediction)[0]
        st.success(f"Prediction: You should {'✅ do' if result == 'Yes' else '🚫 not do'} a Master's.")
```

```python

elif selected == "Graph":
    st.subheader(" Visual Insights from Career Dataset")
    feature = st.selectbox("Choose a feature to visualize", ["GATE_Score", "Salary", "Should_Do_Masters"])

    if feature == "GATE_Score":
        st.subheader(" GATE Score Distribution")
        counts = model_df[feature].value_counts().sort_index()
        labels = [gate_label_map[i] for i in counts.index]

        plt.figure(figsize=(10, 6))
        bars = plt.barh(labels, counts.values, color=plt.cm.PuBuGn(counts.values / max(counts.values)))
        plt.xlabel("Number of Students")
        plt.ylabel("GATE Score")
        plt.title("Horizontal Distribution of GATE Scores")
        st.pyplot(plt.gcf())

    elif feature == "Salary":
        st.subheader(" Salary Spread Across Students")
        plt.figure(figsize=(8, 5))
        plt.boxplot(model_df["Salary"], patch_artist=True,
                    boxprops=dict(facecolor="lightblue", color="blue"),
                    medianprops=dict(color="red"))
        plt.ylabel("Salary (INR)")
        plt.title("Box Plot of Salary")
        st.pyplot(plt.gcf())

    elif feature == "Should_Do_Masters":
        st.subheader(" Master's Decision Breakdown")
        counts = model_df[feature].value_counts().sort_index()
        labels = [master_label_map[i] for i in counts.index]

        plt.figure(figsize=(6, 6))
        wedges, texts, autotexts = plt.pie(counts,
                                           labels=labels,
                                           autopct='%1.1f%%',
                                           startangle=90,
                                           wedgeprops=dict(width=0.4),  # donut effect
                                           colors=["#90EE90", "#FFA07A"])

        plt.setp(autotexts, size=12, weight="bold")
        plt.title("Donut Chart: Should Students Do Masters?")
        st.pyplot(plt.gcf())
```




Streamlit
MastersDecisionSupportSystem.p
Streamlit
localhost:8502
Deploy
Main Menu
Home
About Model
Dataset
Prediction Model
Graph
Masters Decision Support System
Uploaded Dataset (student_career_data.csv)
Student_ID Student_Name Gender Placed GATE_Score Salary Should_Do_Masters
0 S0001 Kunal Male No Good 796246 Yes
1 S0002 Harsh Male No Good 329709 No
2 S0003 Aman Female Yes Poor 1348939 Yes
3 S0004 Ananya Male Yes Poor 794829 No
4 S0005 Kunal Male Yes Poor 696456 No
5 S0006 Nikhil Male No Good 345689 Yes
6 S0007 Aman Female No Good 1198704 Yes
7 S0008 Aman Female No Poor 475037 Yes
8 S0009 Isha Male No Average 508606 No
9 S0010 Ananya Female No Poor 1349458 No


Streamlit
MastersDecisionSupportSystem.p
Streamlit
localhost:8502
Deploy
Main Menu
Home
About Model
Dataset
Prediction Model
Graph
Masters Decision Support System
Predict: Should You Do a Master's?
Select GATE Score
Good
Enter current salary (INR)
1200000
Predict
Prediction: You should not do a Master's.


Streamlit
MastersDecisionSupportSystem.p
Streamlit
localhost:8502
Deploy
Main Menu
Home
About Model
Dataset
Prediction Model
Graph
Masters Decision Support System
Visual Insights from Career Dataset
Choose a feature to visualize
GATE_Score
GATE Score Distribution
Horizontal Distribution of GATE Scores
Poor
Good
Average
GATE Score
0 200 400 600 800 1000 1200 1400
Number of Students

WEEK 2/ Day 11	16 JULY 2024

Objective:

1. Random Forest Project : Fuel consumption
2. Conclusion

1. Random Forest Project : Fuel consumption

Project Overview

- This project aims to build an interactive machine learning-powered Streamlit web application that predicts a vehicle's fuel consumption in city, highway, and combined driving conditions.
- By using real-world vehicle data and user-input features such as engine size, transmission type, fuel type, and emissions, the model helps users estimate fuel efficiency and make informed decisions about driving habits, vehicle selection, and cost planning.

Random Forest Algorithm

- This project uses the **Random Forest Regression** algorithm, a powerful ensemble learning method based on decision trees. A random forest builds **multiple decision trees** during training and outputs the **average of their predictions** in regression tasks, making it highly accurate and resistant to overfitting.

User Input Fields

1. **Vehicle Class** (VEHICLE CLASS)
   This describes the **size or category of the car**.
   - **COMPACT**: Small car like a Swift or Baleno
   - **SUV**: Sports Utility Vehicle (Creta, Scorpio, etc.)
   - **MINICOMPACT**: Very small car
   - **PICKUP TRUCK**: Open-back vehicles for transport
   - **STATION WAGON**: Like a long hatchback with cargo space

   Why it matters: Bigger vehicles usually consume more fuel.

2. **Engine Size** (L) (ENGINE SIZE)
   The size of the engine measured in **litres**. For example:
   - **1.2L** → Small engine (fuel efficient)
   - **2.0L or 3.5L** → Bigger engine (more power, more fuel)

   Why it matters: Larger engines usually consume more fuel but offer more performance.

3. **Cylinders** (CYLINDERS)
   This tells how many **cylinders** the engine has:
   - **4 cylinders** is common for regular cars

- **6 or 8 cylinders** in high-performance cars or SUVs

Why it matters: More cylinders = more fuel consumption, usually.

4. Transmission Type (TRANSMISSION)

This shows how the **gear system** works. The values might look like A6, AM, AV, M5, AS8, etc. Here's what they mean:

Code	Meaning
A	**Automatic** — changes gears automatically.
M	**Manual** — driver changes gears (with clutch).
AS	**Automatic Select Shift** — automatic but lets driver shift manually sometimes.
AV	**Continuously Variable Transmission (CVT)** — smooth automatic gear change without steps.
AM	**Automated Manual** — a manual transmission with automation.
Numbers like 6 or 8	Number of gear speeds (e.g., A6 = 6-speed automatic)

Why it matters: Automatic or CVT systems can affect how efficiently fuel is used.

5. Fuel Type (FUEL)

Type of fuel the car runs on. Typical options:

- **X** = Regular Gasoline (Petrol)
- **Z** = Premium Gasoline (higher performance petrol)
- **D** = Diesel
- **E** = Ethanol (like E85)
- **N** = Natural Gas (CNG)

Why it matters: Different fuels burn differently, affecting fuel consumption.

6. Emissions (g/km) (EMISSIONS)

This is the amount of **carbon dioxide (CO_2)** the car emits per kilometre, in grams.

- **Lower = better for environment**
- Typical range: 100–400 g/km

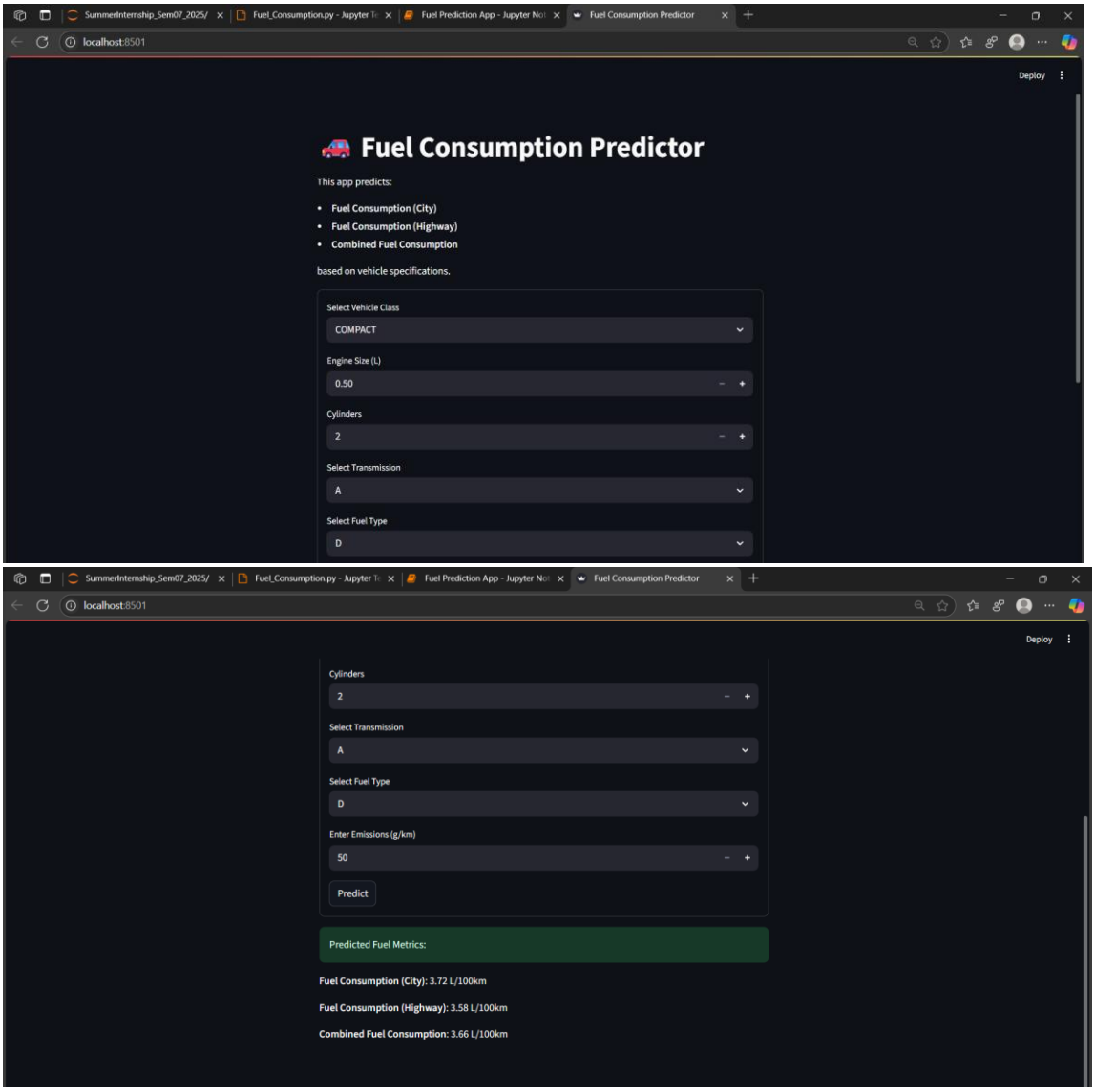
Why it matters: More emissions often mean more fuel used.

Use Case

Scenario	How the output helps
Buying a new/used car	Compare efficiency across vehicles.
Estimating fuel cost	Use these numbers to calculate monthly fuel budget.
Eco-conscious driving	Choose cars or settings that reduce city fuel consumption.

```python
import pandas as pd
import streamlit as st
from sklearn.preprocessing import LabelEncoder
from sklearn.ensemble import RandomForestRegressor
import joblib

st.set_page_config(page_title="Fuel Consumption Predictor", layout="centered")
st.title("🚗 Fuel Consumption Predictor")
st.markdown("""
This app predicts:
- **Fuel Consumption (City)**
- **Fuel Consumption (Highway)**
- **Combined Fuel Consumption**

based on vehicle specifications.
""")

# Load data
data = pd.read_csv("clean_fuel.csv")

# Encode categorical features
class_encoder = LabelEncoder()
transmission_encoder = LabelEncoder()
fuel_encoder = LabelEncoder()

data['VEHICLE CLASS'] = class_encoder.fit_transform(data['VEHICLE CLASS'])
data['TRANSMISSION'] = transmission_encoder.fit_transform(data['TRANSMISSION'])
data['FUEL'] = fuel_encoder.fit_transform(data['FUEL'])

# Features and targets
features = ["VEHICLE CLASS", "ENGINE SIZE", "CYLINDERS", "TRANSMISSION", "FUEL", "EMISSIONS"]
target = ["FUEL CONSUMPTION", "HWY (L/100 km)", "COMB (L/100 km)"]
X = data[features]
y = data[target]
```

```python
@st.cache_resource
def train_model():
    model = RandomForestRegressor(random_state=42)
    model.fit(X, y)
    return model

model = train_model()

# Streamlit inputs
with st.form("prediction_form"):
    vehicle_class = st.selectbox("Select Vehicle Class", class_encoder.classes_)
    engine_size = st.number_input("Engine Size (L)", min_value=0.5, max_value=10.0, step=0.1)
    cylinders = st.number_input("Cylinders", min_value=2, max_value=16, step=1)
    transmission = st.selectbox("Select Transmission", transmission_encoder.classes_)
    fuel = st.selectbox("Select Fuel Type", fuel_encoder.classes_)
    emissions = st.number_input("Enter Emissions (g/km)", min_value=50, max_value=700)
    submitted = st.form_submit_button("Predict")

if submitted:
    try:
        input_data = pd.DataFrame({
            "VEHICLE CLASS": [class_encoder.transform([vehicle_class])[0]],
            "ENGINE SIZE": [engine_size],
            "CYLINDERS": [cylinders],
            "TRANSMISSION": [transmission_encoder.transform([transmission])[0]],
            "FUEL": [fuel_encoder.transform([fuel])[0]],
            "EMISSIONS": [emissions]
        })

        predictions = model.predict(input_data)[0]

        st.success("Predicted Fuel Metrics:")
        st.write(f"**Fuel Consumption (City)**: {predictions[0]:.2f} L/100km")
        st.write(f"**Fuel Consumption (Highway)**: {predictions[1]:.2f} L/100km")
        st.write(f"**Combined Fuel Consumption**: {predictions[2]:.2f} L/100km")

    except ValueError as e:
        st.error(f"Encoding error: {e}")
```



2. Conclusion

During this internship, I gained hands-on experience in data analytics, Power BI, and machine learning. I learned to work with real-time data from APIs, visualize insights using Python libraries and Power BI, and implement regression and classification models for practical use-cases such as COVID analysis, education decision support, and fuel consumption prediction. This journey enhanced my technical skills and problem-solving approach using data-driven methods. The internship has provided a strong foundation for future roles in data science and business intelligence.

Project Repository

🔗 NidhiTank_InfolabzSummerInternship2025-Data-Analytics-ML